

Dynamic Causal Graph Network for Reliable Pipeline Leak Detection[☆]

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ABSTRACT

Achieving reliable leak detection remains a core challenge in pipeline structural health monitoring (SHM), where diagnostic errors can lead to catastrophic failures or costly false alarms that compromise system safety. Multi-sensor SHM combined with Graph Neural Networks (GNNs) offers a promising paradigm for capturing non-Euclidean sensor topologies, yet its reliance on statistical correlations undermines reliability in industrial environments. Confounding factors such as environmental vibrations and fluid turbulence induce spurious correlations among physically disconnected sensors, degrading diagnostic trustworthiness. To enhance detection reliability, this study proposed the Dynamic Causal Graph Network (DCGN), which leveraged causal inference to disentangle spurious connections and recover genuine physical propagation paths. DCGN employs a Causal Node Encoder (CNE) that extracts state embeddings from sensor signals, while the Causal Skeleton Learner (CSL) and Dynamic Causal Mechanism (DCM) collaboratively reconstruct causal connectivity and model time-varying effects using Partial Correlation Momentary Independence (PCMI) priors. A Transformer-based Causal Inference and Reasoning (CIR) module aggregates long-range causal evidence for robust state inference. Experiments on branched and looped pipeline topologies demonstrate that DCGN achieves 96.32% and 96.60% accuracy respectively, significantly outperforming baselines with statistical significance validated by ANOVA. Under extreme conditions including 5-dB signal-to-noise ratio (SNR) and strong impulse noise, DCGN maintains over 86% accuracy, and achieves above 81% accuracy with only 10% training samples, demonstrating strong robustness for reliable operation. Ablation studies confirm the critical contribution of causal modules to performance gains. Causal graph structure validation confirms that the learned skeleton aligns with the physical pipeline topology, and cross-regime transfer experiments demonstrate accuracy degradation within 2.9% across both topologies, further evidencing the distributional invariance of causal representations. DCGN establishes an intelligent diagnostic framework centered on replacing statistical correlation graphs with causal graphs and substituting spurious connections with genuine physical propagation paths, offering a new paradigm for reliable pipeline safety assurance in complex industrial environments.

1. Introduction

Pipelines serve as the backbone of modern industrial infrastructure, transporting water, natural gas, petroleum, and various hazardous chemical fluids across urban and industrial systems [1–5]. Unintended leakage of such fluids can trigger severe safety incidents including toxic exposure and explosions, while simultaneously imposing irreversible contamination on soil, groundwater, and ecosystems [6–11]. Ensuring

the reliable operation of pipeline infrastructure is therefore fundamental to both human safety and environmental protection. Leakage incidents induced by aging infrastructure, environmental corrosion and operational stress have become frequent and unpredictable in modern urban water supply, energy transmission and industrial manufacturing, posing significant reliability challenges [6,12,13]. Pressurized gases and hazardous liquids transported through pipelines are exposed to the natural environment, where harsh conditions and pipe degradation can

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trigger explosions and leaks [14]; for underground pipelines, cathodic protection remains the most effective means against corrosion [15]. Minor leaks, if undetected, can rapidly escalate into catastrophic failures, causing severe economic losses, environmental contamination and public safety threats. The proliferation of sensor networks and Industrial Internet of Things enables real-time acquisition of massive pipeline operational data, establishing the data foundation for intelligent monitoring [16]. Against this background, developing intelligent methods capable of reliably identifying early leak signatures from massive sensor data within the Structural Health Monitoring (SHM) framework has become an urgent need for reliability engineering and infrastructure safety assurance [17–20].

Pipeline leak detection methods have evolved from model-driven approaches relying on physical principles to data-driven paradigms leveraging multi-sensor measurements [21]. Early techniques including pressure transient analysis, acoustic emission and mass balance methods can reveal physical processes of leak occurrence, yet typically require precise system parameters and struggle to maintain reliable performance under complex network topologies, varying operational conditions and environmental disturbances [22]. With the expansion of sensor network deployment and reduction of data acquisition costs, the accumulation of massive industrial data has established the foundation for applying artificial intelligence, particularly deep learning, to pipeline monitoring [23–25]. Leveraging powerful end-to-end representation learning capabilities of Convolutional Neural Networks, Recurrent Neural Networks and attention-based models, these methods have demonstrated remarkable performance in automatic feature extraction and pattern recognition from heterogeneous sensor signals, effectively relaxing the dependence on expert knowledge and manual feature engineering [26–29]. However, traditional deep learning approaches merely concatenate measurements from multiple sensors into vectors or sequences, physically ignoring the spatial relationships of sensors within the pipeline network and their interaction patterns [30, 31]. This limitation compromises the reliability of leak detection, as it fails to capture the essential characteristics of leak wave propagation along physical pipeline paths. In safety-critical SHM applications, false alarms and missed detections carry asymmetric costs, unnecessary shutdowns versus catastrophic escalation, and high overall accuracy alone does not guarantee operational reliability [32]. Simultaneously suppressing both error types is therefore a fundamental requirement for pipeline leak detection.

To overcome this limitation, researchers have begun treating each sensor as a node in a graph structure, constructing inter-sensor connections to describe their interaction patterns [33–35], and subsequently leveraging the powerful representation capability of Graph Neural Networks on non-Euclidean data for pipeline leak detection tasks [36–38]. This perspective of mapping sensors to graph nodes provides a novel pathway for capturing complex topological dependencies within sensor networks. Nevertheless, existing graph neural network methods still face two core challenges. The first is the spatial spurious correlation problem. Most existing methods rely on statistical correlations between sensor measurements to construct graph connections, whereas confounding factors are pervasive in pipeline monitoring scenarios. When leak-induced waves propagate through pipelines, confounding factors such as environmental vibrations, fluid turbulence, and variations in pipe wall materials generate synchronized fluctuations across multiple sensors [39–42], creating strong correlations that are statistical but non-causal. These spurious connections erroneously associate sensor nodes that lack direct physical propagation paths, preventing the model from distinguishing genuine leak wave propagation paths from accompanying responses induced by confounding factors, resulting in substantial performance degradation when confronted with new pipeline topologies or operational conditions. The second is the temporal information attenuation problem. Leak-induced transients exhibit pronounced long-range temporal dependencies, where subtle pressure fluctuations in early time segments

carry critical causal evidence regarding leak source location and type, with these early signals determining the evolution patterns of subsequent sensor responses [43]. However, traditional sequential models suffer from information attenuation across extended temporal horizons [44,45], failing to reliably preserve these cross-temporal causal evidence chains and thereby exhibiting insufficient sensitivity to early weak leak signals.

Researchers have increasingly advocated for integrating causal mechanisms into system health management frameworks to address these challenges [46–50]. Unlike correlation that merely captures statistical co-occurrence, causal inference reveals the true propagation directions of leak disturbances within pipeline networks. Directed edges in causal graphs characterize the directional transmission of leak waves along physical pipelines, while sparse structures reflect the natural constraints of physical pipe connections. Existing causal approaches fall into two categories: statistical methods such as Granger causality and Peter and Clark Momentary Conditional Independence (PCMI) offer strong interpretability but are constrained by linear assumptions, while deep learning methods such as DyGFormer [51–53] enable end-to-end nonlinear modeling yet lack explicit physical constraints. The proposed method embeds PCMI-derived priors into a dynamic GNN architecture, combining the strengths of both. Recent studies demonstrate that causal inference-based methods exhibit superior cross-domain generalization and noise robustness in complex industrial systems [54]. In pipeline leak monitoring scenarios, causal graphs can effectively distinguish synchronized fluctuations induced by confounding factors from physical propagation paths of leak waves, thereby accurately locating leak sources when confronted with unknown pipeline topologies or complex noise interference. Furthermore, dynamic causal graphs allow edge strengths to evolve over time, enabling characterization of energy attenuation and time-varying properties of leak waves during propagation, providing structured support for capturing early weak leak signals. By integrating causal discovery with data-driven learning, such methods promise reliable diagnosis under practical engineering challenges including sensor configuration changes and operational condition drifts.

Building upon this foundation, this work proposes a Dynamic Causal Graph Network (DCGN) that integrates structural causal models with graph neural networks for reliable pipeline leak detection. DCGN comprises four synergistic modules that collectively transform raw multi-sensor observations into causal-informed health state assessments. Causal Node Encoder extracts state embeddings from sensor signals. Causal Skeleton Learner reconstructs sparse causal connectivity constrained by PCMI-derived priors. Dynamic Causal Mechanism generates time-varying causal graphs through intervention-based training. Transformer-based Causal Inference and Reasoning aggregates long-range causal evidence for final state inference. The main contributions are summarized as follows:

1. A joint framework comprising Causal Skeleton Learner and Dynamic Causal Mechanism was designed to disentangle genuine leak wave propagation paths from spurious correlations induced by environmental vibrations and fluid turbulence, enhancing diagnostic reliability in complex pipeline monitoring environments.
2. A Transformer-based Causal Inference and Reasoning module was designed to capture long-range causal dependencies from early subtle pressure fluctuations, enabling reliable detection of incipient leaks before escalation into severe failures.
3. Comprehensive experiments demonstrate that DCGN achieves 96.32% and 96.60% accuracy on branched and looped topologies respectively, maintains robust performance under extreme conditions including 5-dB SNR, 10% training samples, and strong impulse noise, with ablation studies confirming the effectiveness of each causal module.

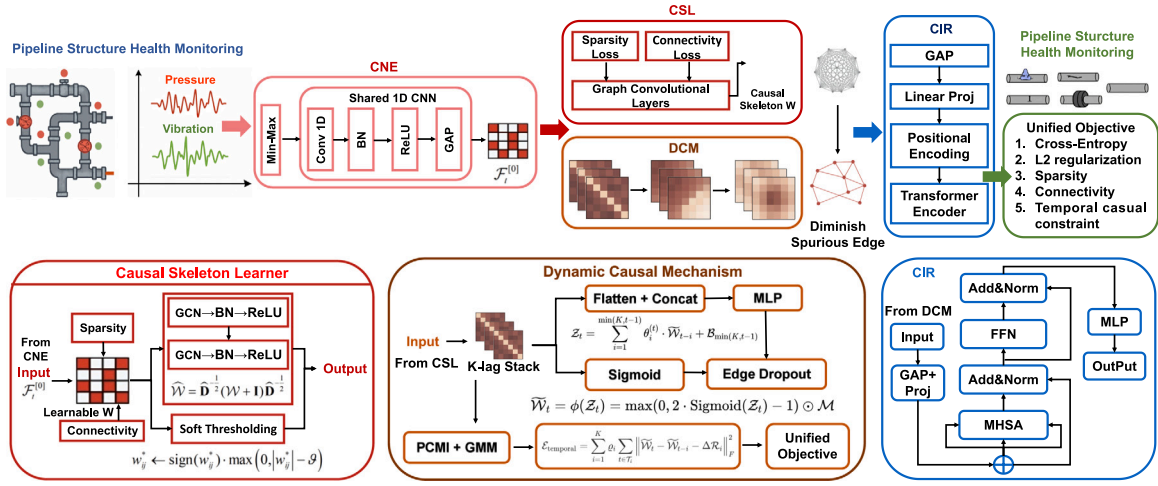


Fig. 1. Overall architecture of the Dynamic Causal Graph Network (DCGN).

Table 1
Summary of key notations.

Notation	Description
n	Number of sensor nodes
T	Sequence length (number of temporal segments)
$\mathcal{W} \in \mathbb{R}^{n \times n}$	Learnable static causal skeleton (adjacency matrix)
$\tilde{\mathcal{W}}_t \in \mathbb{R}^{n \times n}$	Dynamic adjacency matrix at temporal segment t
$\mathcal{F}_t^{[0],q} \in \mathbb{R}^{n \times d_{\text{node}}}$	Node feature matrix at temporal segment t
$\Delta \mathcal{R}_i$	PCMI-derived causal prior for temporal lag i
K	Maximum temporal lag order
$\mathcal{L}_{\text{CE}}$	Cross-entropy classification loss
$\mathcal{E}_{\text{conn}}, \mathcal{E}_{\text{sparse}}, \mathcal{E}_{\text{temporal}}$	Connectivity, sparsity, and temporal causal regularization losses
$\lambda_{\text{conn}}, \beta_{\text{sparse}}, \theta_i, \alpha$	Loss weighting coefficients

2. Proposed methodology

This work formalizes the pipeline leak detection problem as a causal discovery task. Given multi-sensor temporal observations $\{\mathbf{x}_i(t)\}_{i=1}^n$ from n sensors, the objective is to identify the underlying causal structure and dynamic causal mechanisms governing fault propagation. Three assumptions underlie this formulation. First, sensor signals are uniformly sampled at a fixed rate, ensuring consistent temporal resolution across all nodes. Second, additive noise affecting sensor measurements is assumed to be stationary within each observation window, which is consistent with the short-window segmentation strategy adopted in CNE. Third, the causal lag structure among sensors is bounded by a finite maximum lag, reflecting the finite propagation time of leak-induced waves across the pipeline network. Conventional data-driven leak detection methods are frequently influenced by environmental noise and confounding factors, leading to spurious correlations and thereby diminishing the reliability of structural health monitoring (SHM). To address this challenge, this article introduces a Dynamic Causal Graph Network (DCGN), which integrates the principles of structural causal models into the causal inference framework for SHM, enabling reliable health state assessment by disentangling genuine causal pathways from spurious correlations. DCGN builds upon graph neural networks for non-Euclidean sensor topology representation, further integrates the PCMI algorithm for causal discovery and prior extraction, incorporates GMM for causal pattern clustering, and leverages Transformer-based sequence encoding to support long-range temporal reasoning. The DCGN framework comprises four core components: a Causal Node Encoder (CNE) that extracts state embeddings from raw sensor observations, a Causal Skeleton Learner (CSL) that

reconstructs the sparse causal connectivity structure among sensors, a Dynamic Causal Mechanism (DCM) that models time-varying causal effects and incorporates physical priors, and a Causal Inference and Reasoning (CIR) module that performs health state inference based on causal evidence. Key notations are summarized in Table 1, and Fig. 1 illustrates the overall architecture of DCGN.

2.1. Causal node encoder

The primary task in constructing a structural causal model is to explicitly define the observed variables and graph structure within the system. In pipeline health monitoring, this article treats the sensor set $S = \{s_1, s_2, \dots, s_n\}$ as causal observation nodes of the pipeline physical system, where each sensor s_i corresponds to a directly observable random variable representing operational data of the pipeline under specific working conditions and fault states. The signal acquired by each sensor is a one-dimensional time series of length D .

To extract latent representations of node states from these high-dimensional raw waveforms, this article devises a Causal Node Encoder (CNE). CNE adopts a min-max normalization strategy to eliminate dimensional differences among different sensors, and uses the normalized data as graph node feature vectors, thereby uniformly representing multi-sensor signals as temporal graph sequences. Specifically, for the q th sample, its temporal graph representation is defined as:

$$\mathcal{V}^q = \{\mathcal{H}_t^q\}_{t=1}^T, \quad q \in \{1, 2, \dots, Q\} \quad (1)$$

where $\mathcal{H}_t^q = (S, \mathcal{F}_t^{[0],q}, \mathcal{W})$ denotes the graph structure of the t th temporal segment, S is the shared sensor node set, $\mathcal{F}_t^{[0],q} \in \mathbb{R}^{n \times d_{\text{node}}}$ is the node feature matrix prior to entering downstream graph convolutional networks, and $\mathcal{W} \in \mathbb{R}^{n \times n}$ is the global adjacency matrix shared across all samples and temporal segments.

For the signal sequence of sensor s_i in the t th temporal segment, denoted as $\{\mathbf{z}_{i,k}^q\}_{k \in I_t}$, where $\mathbf{z}_{i,k}^q \in \mathbb{R}^d$ represents the raw waveform vector for each time window and I_t is the set of window indices within that temporal segment, the window sequence is first flattened and aggregated to obtain the raw state representation of each sensor in that temporal segment:

$$\mathbf{r}_{i,t}^q = \text{vec} \left(\left[\mathbf{z}_{i,k}^q \right]_{k \in I_t} \right) \in \mathbb{R}^{d_{\text{raw}}} \quad (2)$$

where d_{raw} denotes the dimensionality of the flattened one-dimensional feature.

The CNE module employs a parameter-shared one-dimensional convolutional waveform encoder to extract robust pipeline health state embeddings from raw observations. The raw waveform vector $\mathbf{r}_{i,t}^q$ of each

sensor is fed into a multi-layer one-dimensional convolutional network, followed by batch normalization, nonlinear activation, and Global Average Pooling (GAP) to compress the features into a fixed-length vector:

$$\mathbf{f}_{i,t}^{[0],q} = \text{GAP} \left(\sigma \left(\text{BN} \left(\mathbf{W}_{\text{enc}} * \mathbf{r}_{i,t}^q + \mathbf{b}_{\text{enc}} \right) \right) \right) \in \mathbb{R}^{d_{\text{node}}} \quad (3)$$

where $\mathbf{W}_{\text{enc}}$ and $\mathbf{b}_{\text{enc}}$ are the learnable parameters of the one-dimensional convolutional encoder, $\text{BN}(\cdot)$ denotes the batch normalization operator, $\sigma(\cdot)$ represents the nonlinear activation function, and $\text{GAP}(\cdot)$ denotes the global average pooling operation. Through this approach, CNE shares parameters across all sensors and temporal segments, a design predicated on the assumption that different nodes exhibit similar responses to identical types of perturbations and adhere to universal physical laws.

The causal node state embedding matrix for the t th temporal segment is obtained by concatenating the embeddings of individual nodes:

$$\mathcal{F}_t^{[0],q} = [(\mathbf{f}_{1,t}^{[0],q})^\top, (\mathbf{f}_{2,t}^{[0],q})^\top, \dots, (\mathbf{f}_{n,t}^{[0],q})^\top]^\top \in \mathbb{R}^{n \times d_{\text{node}}} \quad (4)$$

Sample $\mathcal{U}^q$ is modeled as a temporal graph sequence comprising multiple temporal segment graphs $\{\mathcal{H}_t^q\}$. The downstream task performs graph-level classification on these temporal graphs to output the corresponding pipeline operational state. For notational simplicity, the sample subscript q is omitted hereafter when no ambiguity arises.

Thus far, CNE has accomplished the state embedding of observed variables in the pipeline system. The next section elaborates on the Causal Skeleton Learner (CSL) and unveils the physical connectivity relationships among these nodes.

2.2. Causal skeleton learner

Urban water supply pipeline networks inherently possess physical connectivity relationships that govern the causal dependencies among nodes. However, directly learning inter-node dependencies from observational data is susceptible to environmental noise and confounding factors, leading models to capture spurious correlations rather than genuine causal pathways. To address this challenge, this article devises a Causal Skeleton Learner (CSL), which aims to reconstruct the weighted causal skeleton of the system from observational data—specifically, the sparse graph structure representing authentic physical causal connections.

CSL shares a learnable global adjacency matrix $\mathcal{W} \in \mathbb{R}^{n \times n}$ across all samples and temporal segments, where the matrix element w_{ij} quantifies the causal connection strength between sensors s_i and s_j . It is worth noting that leak-induced waves propagate as elastic waves bidirectionally through both the fluid medium and the pipe wall, rendering the physical coupling between any two sensors inherently symmetric. Accordingly, a symmetrization operation $\mathcal{W} \leftarrow (\mathcal{W} + \mathcal{W}^\top)/2$ is applied to the learned adjacency matrix, and self-loops are explicitly removed by setting diagonal elements to zero to enforce this physical prior, yielding a symmetric $\mathcal{W}$ without self-loops. To ensure mathematical compatibility between the connectivity loss and the soft-thresholding mechanism while generating genuine physical sparsity, CSL adopts a two-stage parameterization scheme: a raw parameter matrix $\tilde{\mathcal{W}} \in \mathbb{R}^{n \times n}$ is maintained, soft-thresholding acts on $\tilde{\mathcal{W}}$, and the adjacency matrix $\mathcal{W}$ is obtained by applying a zero-centered shifted Sigmoid transformation as $\mathcal{W} = \max(0, 2 \cdot \text{Sigmoid}(\tilde{\mathcal{W}}) - 1)$. When the raw parameters are truncated to 0 by soft-thresholding, this mapping strictly guarantees that the final edge weights are 0, while smoothly constraining positive parameters to the interval $(0, 1)$, which, together with a small constant, guarantees that the node degree vector $\mathbf{d} = \mathcal{W}\mathbf{1}$ is strictly positive and the logarithmic operation in the connectivity loss is always well-defined. To eliminate redundant non-causal connections while ensuring the integrity of fault propagation pathways, this article introduces two regularization losses to optimize the learning process of the causal skeleton.

Sparsity Loss: Grounded in the faithfulness assumption of causal inference, genuine physical causal pathways are typically sparse, whereas

redundant connections often originate from environmental noise or indirect pathways. The sparsity loss penalizes excessive edge weights and overly dense connectivity patterns through the squared Frobenius norm of the adjacency matrix, thereby guiding the model to eliminate spurious correlations and retain authentic causal connections. Specifically, the sparsity loss is defined as:

$$\mathcal{E}_{\text{sparse}}(\mathcal{W}) = \|\mathcal{W}\|_F^2 = \sum_{i=1}^n \sum_{j=1}^n w_{ij}^2 \quad (5)$$

This loss term imposes stronger penalties on larger edge weights, facilitating the elimination of spurious correlations and ultimately enhancing the accuracy of the causal skeleton and the reliability of the system.

Connectivity Loss: The connectivity loss ensures that faults, as system perturbations, can identify at least one valid propagation pathway within the causal graph. To maintain the connectivity of the causal skeleton, the degree vector of nodes is defined as $\mathbf{d} = \mathcal{W}\mathbf{1}$, where $\mathbf{1}$ denotes the all-ones vector, and the following connectivity loss is employed:

$$\mathcal{E}_{\text{conn}}(\mathcal{W}) = - \sum_{i=1}^n \log(d_i + \epsilon) \quad (6)$$

where ϵ is a small constant preventing logarithmic singularity. This loss term penalizes nodes with degrees approaching zero, thereby avoiding isolated nodes and ensuring the structural integrity of the causal graph.

Based on the learned causal skeleton $\mathcal{W}$ (in the complete dynamic model, GCN actually utilizes the dynamic adjacency matrices $\tilde{\mathcal{W}}_t$ generated by DCM), CSL introduces two cascaded Graph Convolutional Layers (GCN) to propagate node state information along causal pathways. Specifically, for the t th temporal segment, the input to the graph convolutional layers is the node feature matrix $\mathcal{F}_t^{[0]}$ output by CNE, and the update process of the graph convolutional layers is as follows:

$$\mathcal{F}_t^{[1]} = \sigma \left(\hat{\mathcal{W}} \mathcal{F}_t^{[0]} \boldsymbol{\Theta}^{[0]} \right) \quad (7)$$

$$\mathcal{F}_t^{[2]} = \sigma \left(\hat{\mathcal{W}} \text{Dropout}(\mathcal{F}_t^{[1]}) \boldsymbol{\Theta}^{[1]} \right) \quad (8)$$

$$\hat{\mathcal{W}} = \hat{\mathbf{D}}^{-\frac{1}{2}} (\mathcal{W} + \mathbf{I}) \hat{\mathbf{D}}^{-\frac{1}{2}} \quad (9)$$

Batch normalization and nonlinear activation functions ($\sigma(\cdot)$) are applied after each layer to enhance training stability and expressive capacity.

Soft-Thresholding Denoising Mechanism: To further filter weak spurious connections introduced by noise, CSL applies a soft-thresholding denoising mechanism to the raw parameter matrix $\tilde{\mathcal{W}}$ after each parameter update. Specifically, let $\tilde{\mathcal{W}}^*$ denote the raw parameters after gradient update; each matrix element undergoes the following operation:

$$\tilde{w}_{ij}^* \leftarrow \text{sign}(\tilde{w}_{ij}^*) \cdot \max \left(0, \left| \tilde{w}_{ij}^* \right| - \vartheta \right) \quad (10)$$

where ϑ is the soft-threshold coefficient, simulating the energy threshold of signal propagation in physical systems. Since soft-thresholding truncates raw parameter scores with small magnitudes to zero, combined with the shifted Sigmoid mapping $\mathcal{W} = \max(0, 2 \cdot \text{Sigmoid}(\tilde{\mathcal{W}}) - 1)$, the adjacency matrix always remains non-negative and can achieve genuine sparsity, and the logarithmic operation in the connectivity loss is well-defined throughout training. This operation is equivalent to imposing an L_1 -type sparsity constraint that retains only critical pathways whose causal influence strength exceeds the physical threshold, thereby significantly enhancing the robustness of the causal skeleton.

2.3. Dynamic causal mechanism

The static causal skeleton $\mathcal{W}$ cannot effectively describe the dynamic propagation process of leak-induced waves in pipeline networks. In physical systems, although causal structures such as pipeline topology typically remain relatively stable, the strength of causal effects, such

as pressure wave propagation and attenuation, exhibits time-varying characteristics as operating conditions change. Furthermore, the causal skeleton learned by CSL may still contain spurious correlations introduced by environmental noise and confounding factors.

To construct a trustworthy causal model, it is imperative to introduce a rigorous causal discovery mechanism to constrain the temporal evolution process and prevent the model from overfitting to statistical biases during training. To this end, this article devises a Dynamic Causal Mechanism (DCM), which integrates data-driven causal priors with a recursive generation mechanism to simulate the flow and attenuation of physical energy within the causal skeleton.

DCM employs the global adjacency matrix $\mathcal{W}$ learned by CSL as the initial causal skeleton and recursively generates a sequence of dynamic adjacency matrices $\{\tilde{\mathcal{W}}_1, \dots, \tilde{\mathcal{W}}_T\}$ across the temporal dimension, where the first time step retains the static skeleton:

$$\tilde{\mathcal{W}}_1 = \mathcal{W} \quad (11)$$

For subsequent time steps $t > 1$, DCM leverages the adjacency matrices from the previous K historical time steps (where K denotes the maximum temporal lag) to characterize the dynamic evolution of causal effects. To achieve this, DCM employs a multi-layer perceptron (MLP) to implement a temporal weight adaptation module, thereby modeling the time-varying characteristics of causal strength. When generating the adjacency matrix for the t th time step, the preceding $\min(K, t-1)$ historical adjacency matrices $\{\tilde{\mathcal{W}}_{t-1}, \dots, \tilde{\mathcal{W}}_{t-\min(K, t-1)}\}$ are first flattened and concatenated into a one-dimensional vector, which serves as input to the temporal MLP. The MLP outputs a set of dynamic modulation coefficients $\{\Delta\theta_1^{(t)}, \dots, \Delta\theta_K^{(t)}\}$, which are combined with the base learnable temporal weights $\{\theta_1, \dots, \theta_K\}$ to yield updated temporal weights:

$$\theta_i^{(t)} = \theta_i \cdot \left(1 + \Delta\theta_i^{(t)}\right), \quad i = 1, \dots, K \quad (12)$$

The linear combination of the adjacency matrix prior to activation at the t th time step can be expressed as:

$$\mathcal{Z}_t = \sum_{i=1}^{\min(K, t-1)} \theta_i^{(t)} \cdot \tilde{\mathcal{W}}_{t-i} + \mathcal{B}_{\min(K, t-1)} \quad (13)$$

where $\mathcal{B}_{\min(K, t-1)} \in \mathbb{R}^{n \times n}$ is a learnable bias matrix corresponding to the effective historical steps.

To enhance model robustness against sensor failures or data miss- ingness, DCM introduces an intervention training strategy. In this strategy, two-dimensional Dropout is applied to $\mathcal{Z}_t$, generating a Bernoulli-distributed mask matrix $\mathcal{M} \in \{0, 1\}^{n \times n}$ that randomly drops a subset of edge connections during training, simulating soft interventions on causal pathways. Subsequently, edge weights are constrained to a bounded interval via the shifted Sigmoid function:

$$\tilde{\mathcal{W}}_t = \phi(\mathcal{Z}_t) = \max(0, 2 \cdot \text{Sigmoid}(\mathcal{Z}_t) - 1) \odot \mathcal{M} \quad (14)$$

This intervention mechanism compels the model to avoid relying on a single causal pathway for inference; instead, it learns multiple redundant causal propagation channels, thereby substantially enhancing system robustness in real-world deployment.

To prevent DCM from learning evolution patterns inconsistent with physical laws, this work introduced a data-driven causal discovery mechanism to construct causal prior constraints. The Partial Correlation Momentary Independence PCMI algorithm was employed to extract essential causal connectivity patterns across different temporal lags from training data. PCMI proceeds in two steps. The first step applies the partial correlation-based PC algorithm at significance level 0.05 to identify the candidate parent node set of each target node. The second step performs the MCI test on each candidate connection to confirm genuine causal relationships, where the conditioning set Z for source sensor node X and target sensor node Y is defined as the union of the target node's causal parent set excluding X and the source node's parent set, thereby simultaneously blocking confounding backdoor paths

and indirect pathways. The MCI test employs the partial correlation coefficient as the test statistic, performing a Pearson correlation test on the linear residuals of X and Y conditioned on Z , and connections with p-values below 0.05 are identified as genuine causal links, thereby removing spurious connections induced by confounders or indirect pathways.

For each temporal lag $i \in \{1, \dots, K\}$, PCMI computes the causal influence strength from source sensor X to target sensor Y conditioned on set Z :

$$\text{PCMI}_{X \rightarrow Y|Z}^{(i)} = I(X_{t-i}; Y_t | Z_{t-i}) - I(X_{t-i}; Y_t | Z_t) \quad (15)$$

where $I(\cdot; \cdot | \cdot)$ denotes conditional mutual information. The first term, $I(X_{t-i}; Y_t | Z_{t-i})$, computes conditional mutual information based on the historical conditioning set Z_{t-i} , capturing all potential dependencies. The second term, $I(X_{t-i}; Y_t | Z_t)$, conditions on the current set Z_t , effectively blocking backdoor paths induced by environmental confounders through the introduction of instantaneous observations. The difference between these two terms quantifies the genuine causal influence strength from X_{t-i} to Y_t , thereby identifying intrinsic causal mechanisms unaffected by environmental variations. It should be noted that the directionality in PCMI ($X_{t-i} \rightarrow Y_t$) characterizes the temporal lag order of causality rather than the spatial topological direction between sensors. The symmetrization of $\mathcal{W}$ applies exclusively to the spatial graph structure, while the PCMI prior constrains the temporal evolution direction of the dynamic matrix difference $\tilde{\mathcal{W}}_t - \tilde{\mathcal{W}}_{t-i}$. These two aspects operate at distinct levels and do not conflict.

Based on the PCMI algorithm, for each temporal lag i , causal connection matrices $\{C_i^{(q)}\}$ were collected from all samples and representative causal patterns $\Delta\mathcal{R}_i$ were extracted via Gaussian Mixture Model(GMM) clustering, selecting the mean of the component with the largest weight as the cluster center. These PCMI-derived priors serve as fixed parameters during training, excluded from gradient updates, and are employed to constrain the learning direction of DCM, preventing overfitting to statistical biases. The linear assumption underlying PCMI is physically justified in this context. Leak-induced pressure and elastic waves propagate through pipelines predominantly via linear elastic mechanisms, and the inter-sensor interactions within short temporal segments are adequately represented by linear regression. The PCMI priors are computed offline over the training set only once before training commences and are never recomputed during training iterations. The computational complexity is $O(N \cdot n^2 \cdot K)$, where N denotes the number of training samples, $n = 4$ is the number of sensor nodes, and $K = 3$ is the maximum temporal lag order, and the negligibly small n renders the overall computational overhead insignificant.

The temporal causal loss is defined as:

$$\mathcal{E}_{\text{temporal}} = \sum_{i=1}^K \rho_i \sum_{t \in \mathcal{T}_i} \left\| \tilde{\mathcal{W}}_t - \tilde{\mathcal{W}}_{t-i} - \Delta\mathcal{R}_i \right\|_F^2 \quad (16)$$

where $\mathcal{T}_i$ denotes the set of temporal indices available for lag i (e.g., $t \in \{i+1, \dots, T\}$), and $\rho_i > 0$ represents the corresponding weighting coefficient. This loss term encourages the adjacency matrix differences $\tilde{\mathcal{W}}_t - \tilde{\mathcal{W}}_{t-i}$ learned by DCM to statistically approximate the causal prior patterns $\Delta\mathcal{R}_i$ extracted by PCMI, thereby reinforcing consistency with genuine physical processes while maintaining model flexibility. By integrating physically consistent causal priors with dynamic evolution mechanisms, DCM constructs a dynamic causal graph sequence $\{\tilde{\mathcal{W}}_t\}$ that adheres to dynamical laws and exhibits data adaptability, thereby establishing a solid foundation for subsequent causal pathway-based fault inference.

2.4. Causal inference and reasoning

In the preceding modules, causal node state embeddings (CNE), the sparse causal skeleton (CSL), and the dynamic causal graph sequence (DCM) had been constructed. To render a final safety assessment, a

causal reasoning mechanism was required that could integrate these dynamically evolving causal evidence and perform fault inference based on causal evidence.

Pipeline leak signals typically exhibit pronounced long-range dependency characteristics: subtle anomalous waveforms in early temporal segments may carry critical causal information, while prominent anomalies in later segments often result from the cumulative temporal propagation of these early causal perturbations. Traditional Long Short-Term Memory (LSTM) networks, constrained by their sequential processing mechanism, suffer from information attenuation as hidden states propagate across multiple time steps, thus failing to effectively preserve causal evidence between distant temporal segments.

To address this challenge, this article devises a Causal Inference and Reasoning (CIR) module. Leveraging the self-attention mechanism of Transformer encoders, CIR directly computes causal association strengths between arbitrary time steps within the dynamic causal graph sequence, thereby effectively capturing causal effect propagation patterns spanning long temporal horizons.

CIR first extracts the causal context representation of each temporal segment. Specifically, for the i th temporal segment, let $\mathcal{F}_i^{[2]} \in \mathbb{R}^{n \times d_{\text{hidden}}}$ denote the node feature matrix output by the second graph convolutional layer of CSL, where the i th row corresponds to the feature vector of node s_i at temporal segment i , denoted as $\mathbf{f}_{i,i}^{[2]} \in \mathbb{R}^{d_{\text{hidden}}}$. CIR aggregates node features into a graph-level causal context vector via global average pooling (GAP):

$$\mathbf{h}_i = \frac{1}{n} \sum_{i=1}^n \mathbf{f}_{i,i}^{[2]} \in \mathbb{R}^{d_{\text{hidden}}} \quad (17)$$

For all temporal segments of a single sample, the causal context sequence $\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_T] \in \mathbb{R}^{T \times d_{\text{hidden}}}$ was obtained, where each $\mathbf{h}_i$ encodes the global causal state at time i .

Next, CIR maps the causal context sequence to the Transformer model dimension d_{model} via a linear projection layer and incorporates positional encoding to preserve temporal ordering information:

$$\mathbf{H}^{[0]} = \mathbf{H}\mathbf{W}_{\text{proj}} + \mathbf{PE} \quad (18)$$

where $\mathbf{W}_{\text{proj}} \in \mathbb{R}^{d_{\text{hidden}} \times d_{\text{model}}}$ is a learnable projection matrix, and $\mathbf{PE} \in \mathbb{R}^{T \times d_{\text{model}}}$ is the sinusoidal positional encoding matrix. Subsequently, a multi-layer Transformer encoder is constructed, which directly computes causal association weights between arbitrary pairs of time steps via the multi-head self-attention mechanism, capturing long-range causal dependencies. Each layer integrates feed-forward networks, residual connections, and layer normalization to enhance training stability, ultimately outputting the hidden state $\mathbf{H}^{[N_{\text{layer}}]} \in \mathbb{R}^{T \times d_{\text{model}}}$.

CIR extracts the vector $\mathbf{h}_T^{[N_{\text{layer}}]}$ from the final time step as the global representation of the entire causal evolution process. This vector integrates causal contextual information across the full temporal span, encompassing the complete causal chain from the initial state to the current state.

In the health state assessment phase, CIR employs a two-layer fully connected network (MLP) to apply nonlinear transformations and Dropout regularization to $\mathbf{h}_T^{[N_{\text{layer}}]}$, subsequently mapping it to the final health state category space to output predictions for different leak types and normal states. Since this decision relies on global causal context rather than localized temporal signals, CIR exhibits more pronounced advantages in noise robustness and trustworthiness compared to conventional purely data-driven methods. During training, cross-entropy loss supervises the predicted probability distribution, thereby achieving health state inference based on causal evidence.

2.5. Unified optimization objective

The training process of DCGN constitutes a multi-objective optimization problem aimed at balancing monitoring accuracy with the integrity of the causal structure. Monitoring accuracy is ensured through

supervised loss to enable the model to accurately identify the health state of the pipeline system, while the integrity of the causal structure is guaranteed through regularization constraints to ensure that the learned causal graph adheres to physical laws, thereby enhancing system reliability and trustworthiness. Let Φ denote the complete set of learnable parameters of the model; the overall optimization objective is defined as:

$$J_{\text{total}} = \mathcal{L}_{\text{CE}}(\mathbf{y}, \hat{\mathbf{y}}) + \alpha \|\Phi\|_2^2 + \lambda_{\text{conn}} \mathcal{E}_{\text{conn}}(\mathcal{W}) + \beta_{\text{sparse}} \mathcal{E}_{\text{sparse}}(\mathcal{W}) + \mathcal{E}_{\text{temporal}} \quad (19)$$

where $\mathcal{L}_{\text{CE}}(\mathbf{y}, \hat{\mathbf{y}})$ represents the cross-entropy loss for health state classification, ensuring monitoring accuracy; $\alpha \|\Phi\|_2^2$ is the L_2 regularization term, employed to prevent overfitting; $\lambda_{\text{conn}} \mathcal{E}_{\text{conn}}(\mathcal{W})$ and $\beta_{\text{sparse}} \mathcal{E}_{\text{sparse}}(\mathcal{W})$ respectively ensure the connectivity and sparsity of the static causal skeleton $\mathcal{W}$ in CSL, maintaining the physical rationality of the causal structure; $\mathcal{E}_{\text{temporal}}$ represents the temporal causal loss defined by DCM based on the dynamic adjacency matrices $\{\tilde{\mathcal{W}}_i\}$ and PCMI priors $\{\Delta \mathcal{R}_i\}$, constraining the dynamic causal evolution process to conform to genuine physical mechanisms.

Through joint optimization of these objectives, DCGN successfully constructs a physically consistent and trustworthy causal model while guaranteeing high monitoring accuracy, thereby providing a reliable solution for health state assessment of pipeline systems. The weighting coefficients λ_{conn} , β_{sparse} , and ϕ_i of each loss term are determined through systematic experiments, and no dynamic balancing mechanism is required among the loss components. DCGN training runs for the full number of epochs without early stopping, and the checkpoint achieving the highest validation accuracy within each fold of the five-fold cross-validation is selected as the final result. The benchmark dataset employed in this study contains balanced sample counts across all fault categories, such that $\mathcal{L}_{\text{CE}}$ provides fair training for each class without the need for class-weighted loss. The complete CSL $\leftrightarrow$ DCM training procedure is summarized in Algorithm 1.

3. Experimental results and analysis

3.1. Datasets and experimental setup

This work employs the pipeline leak detection benchmark dataset released by Aghashahi et al. [55], which was acquired from a laboratory-scale water distribution network testing platform. The sensor array comprises four sensors, namely two piezoelectric accelerometers PCB 333B50 and two dynamic pressure sensors PCB 102B16, synchronized via two NI-9234 acquisition modules at a sampling rate of 51.2 kHz, with each experiment recording 30 s of continuous measurements. Specifically, the accelerometers feature ceramic sensing elements with a resonant frequency of at least 20 kHz and a frequency range of 0.5 to 3000 Hz, while the pressure sensors feature quartz sensing elements with a measurement resolution of 0.007 kPa and a resonant frequency of at least 500 kHz. The dataset encompasses both Branched and Looped pipeline network topologies and provides measurements under various hydraulic conditions, including zero flow, 0.18 L/s steady flow, and 0.47 L/s steady flow. This study conducts experiments on both topologies under a background flow of 0.18 L/s, with DCGN trained and evaluated independently for each topology. In addition, generalization experiments across multiple flow regimes and both topology configurations are further conducted to assess cross-condition and cross-topology transferability. Regarding pipeline leak types, the data for each operating condition uniformly cover five operational states: orifice leak, longitudinal crack leak, circumferential crack leak, gasket leak, and no-leak. Each leak type is induced independently on a separate pipe segment; label definitions are detailed in Table 2. Furthermore, the raw signals contain no missing values, and min-max normalization is applied to each sensor channel to eliminate dimensional discrepancies.

Algorithm 1 CSL $\leftrightarrow$ DCM Joint Training in DCGN

```

1: Pre-training: offline PCMI prior computation (once per fold, on
   current-fold training set only)
2: for  $i = 1$  to  $K$  do
3:   compute  $C_i^{(q)}$  via PCMI on training set
4:   fit GMM
5:    $\Delta R_i \leftarrow$  mean of highest-weight component  $\triangleright$  fixed, no grad
6: end for
7: Training loop
8: for  $epoch = 1$  to  $E$  do
9:   for each mini-batch do
10:    CSL: build static skeleton
11:     $W \leftarrow \max(0, 2 \cdot \text{Sigmoid}(\overline{W} - 1))$   $\triangleright$  two-stage
       parameterization
12:     $W \leftarrow \frac{W+W^T}{2}$   $\triangleright$  symmetry prior
13:     $W \leftarrow W \odot (1 - I)$   $\triangleright$  remove self-loops
14:    DCM: dynamic graphs  $\leftarrow$  initialized from CSL output  $W$ 
15:     $\tilde{W}_1 \leftarrow W$ 
16:    for  $t = 2$  to  $T$  do
17:       $\{\theta_t^i\} \leftarrow \theta_i \cdot (1 + \text{MLP}([\tilde{W}_{t-1}, \dots, \tilde{W}_{t-K}]))$ 
18:       $Z_t \leftarrow \sum_i \theta_t^i \cdot \tilde{W}_{t-i} + B_{\min(K,t-1)}$ 
19:       $\tilde{W}_t \leftarrow \max(0, 2 \cdot \text{Sigmoid}(Z_t) - 1) \odot \text{Bernoulli}(p)$   $\triangleright$  soft
       intervention
20:    end for
21:    CNE + CSL GCN: node embeddings
22:    for  $t = 1$  to  $T$  do
23:       $\hat{W}_t \leftarrow \hat{D}_t^{-1/2}(\tilde{W}_t + I)\hat{D}_t^{-1/2}$   $\triangleright$  normalized dynamic
       adjacency
24:       $F_t^{[2]} \leftarrow \text{GCN}(\text{CNE}(\text{inputs}_t), \hat{W}_t)$ 
25:    end for
26:    CIR: classification
27:     $\hat{y} \leftarrow \text{CIR}(\{F_t^{[2]}\})$ 
28:    Joint loss
29:     $J \leftarrow L_{\text{CE}} + \lambda_{\text{conn}} \cdot E_{\text{conn}}(W) + \beta_{\text{sparse}} \cdot E_{\text{sparse}}(W)$ 
30:       $+ \sum_i \rho_i \sum_t \|\tilde{W}_t - \tilde{W}_{t-i} - \Delta R_i\|_F^2 + \alpha \|\Phi\|^2$ 
31:    Backward + CSL post-update (no gradient)
32:     $\Phi \leftarrow \Phi - \eta \cdot \frac{\partial J}{\partial \Phi}$ 
33:     $\overline{W} \leftarrow \text{sign}(\overline{W}) \cdot \max(0, |\overline{W}| - \vartheta)$   $\triangleright$  soft-thresholding on  $\overline{W}$ 
34:  end for
35: end for

```

The dataset is partitioned according to a five-fold cross-validation strategy to ensure robustness of the training and test sets and to obtain reliable generalization performance evaluation. Crucially, the PCMI priors are recomputed independently for each fold using only the training subset of that fold, with the corresponding validation/test data strictly excluded, thereby preventing any data leakage between prior estimation and model evaluation. Training of the DCGN model is implemented on Ubuntu 22.04 operating system, Python 3.11 development environment, and PyTorch deep learning framework, with computational acceleration provided by an NVIDIA GeForce RTX 4070 graphics card and development conducted using Visual Studio Code 1.84. Model training employs accuracy as the primary evaluation metric, with final model performance measured by the average accuracy across five-fold cross-validation. The training hyperparameter configuration for the DCGN model is detailed in Table 3, and the key implementation settings of each module are summarized in Table 4.

3.2. Hyperparameter sensitivity analysis

To validate the stability and adaptability of the DCGN graph classification framework in pipeline leak detection tasks, this section systematically evaluates seven categories of key hyperparameters under

Table 2

Fault mode classification labels.

Fault mode	Label
Orifice leak	0
Longitudinal crack	1
Circumferential crack	2
Gasket leak	3
No leak	4

Table 3

Hyperparameter configuration for DCGN training.

Parameter	Value
Batch size	32
Epochs	600
Learning rate	1e-3
Optimizer	Adam
Dropout	0.3

Table 4

Key implementation settings of DCGN modules.

Module	Parameter	Value
CNE	Conv1d layers	2
CNE	Window length ω	1024
CNE	Node embedding dim d_{node}	64
CNE	Activation	ReLU
CNE	Normalization	Batch normalization
CSL	GCN layers	2
CSL	Hidden dim d_{hidden}	128
CSL	Activation	ReLU
CSL	Regularization	Dropout ($p = 0.5$)
DCM	Max temporal lag K	3
DCM	Model dim	128
CIR	Transformer layers N_{layer}	2
CIR	Model dim d_{model}	128
CIR	Attention heads	4
CIR	FFN dim	256
CIR	Activation	GELU
CIR	Regularization	Dropout ($p = 0.3$), Layer normalization

Branched and Looped pipeline topologies, with results presented in Fig. 2. Subfigures (a) and (b) present the analysis of training optimizers, causal prior clustering methods, and regularization coefficients λ and β under Branched and Looped topologies, respectively. All four optimizers including Adam, AdamW, SGD, and Adagrad yield graph classification accuracy exceeding 95% under both topologies, with Adam achieving the best performance, confirming robustness to optimizer selection. GMM achieves the highest accuracy under both topologies, substantially outperforming DBSCAN and K-means, attributable to its capacity to flexibly capture multimodal causal pattern distributions among graph node sensors through multiple Gaussian components. Regarding regularization coefficients, both topologies reach peak graph classification performance at $\lambda = 0.1$ and $\beta = 0.01$, where the optimal balance between sparsity and connectivity constraints on the causal skeleton effectively eliminates spurious connections.

Subfigures (c) and (d) present sensitivity results for the causal loss coefficient and input signal length, respectively. As the causal loss coefficient increases from 2 to 3, graph classification accuracy peaks under both topologies, indicating that moderate causal constraints effectively guide the model to learn physically consistent leak wave propagation dynamics along the pipeline graph structure; further increasing the coefficient to 4 and 5 leads to accuracy degradation, as excessively strong constraints suppress the graph neural network's adaptive learning. Across five signal length settings from $L = 1024$ to $L = 5120$, accuracy remains stable between 96% and 98% under both topologies, demonstrating robustness to signal length variations. Subfigures (e), (f), and (g) further analyze three PCMI parameters that directly govern causal prior quality, namely the MCI test p -value threshold, the number of GMM components, and the maximum temporal lag K . A p -value

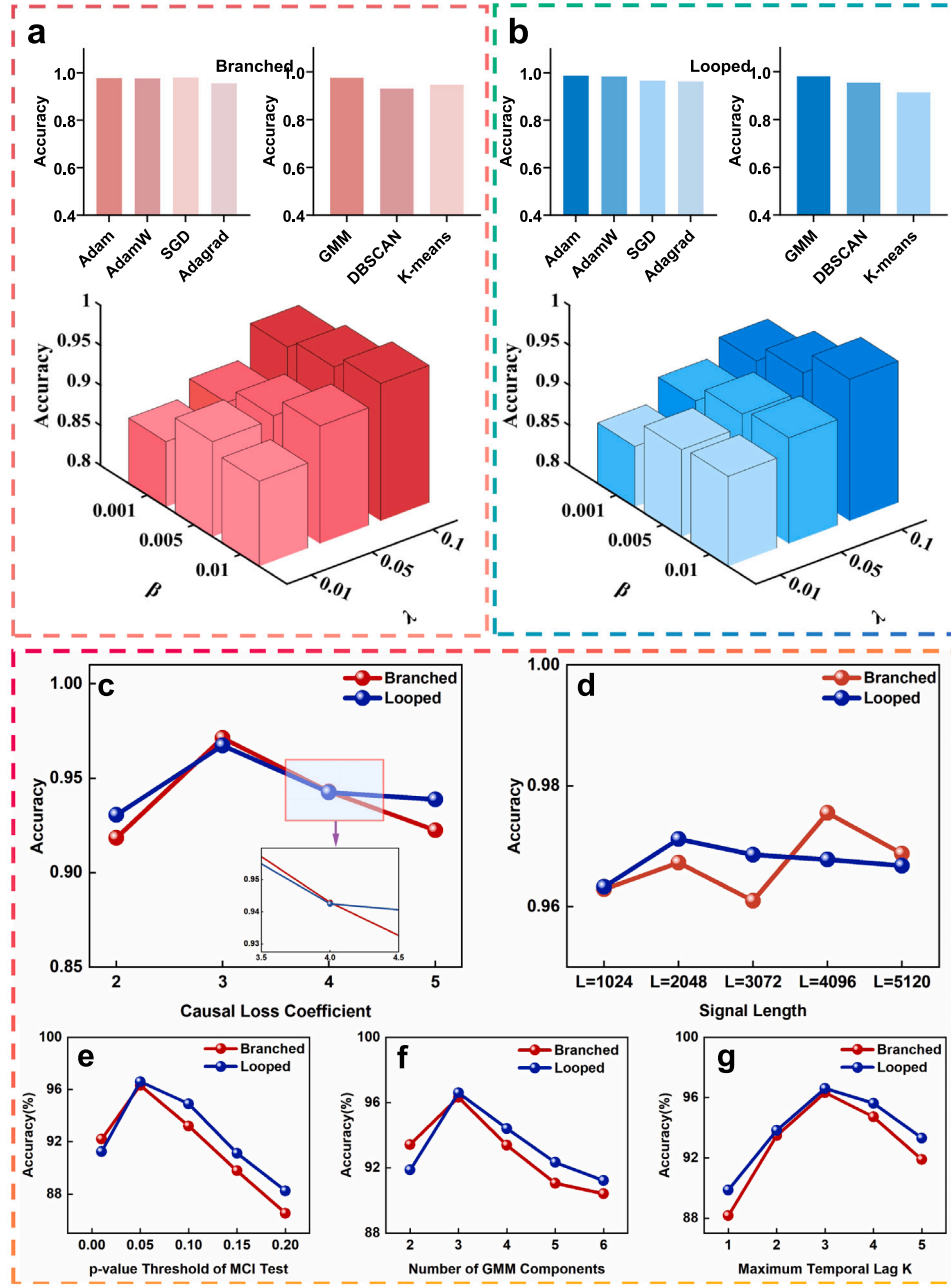


Fig. 2. Hyperparameter sensitivity analysis of DCGN for pipeline leak detection. Subfigure (a) presents results under the Branched topology for training optimizer selection, causal prior clustering method, and the joint effect of regularization coefficients λ and β on causal skeleton learning. Subfigure (b) presents the corresponding results under the looped topology. Subfigure (c) shows the sensitivity of the causal loss coefficient under both topologies. Subfigure (d) shows the effect of input signal length on graph classification accuracy under both topologies. Subfigure (e) analyzes the influence of the MCI test p -value threshold on detection performance. Subfigure (f) examines the effect of the number of GMM components on detection accuracy. Subfigure (g) illustrates the sensitivity of the maximum temporal lag K on detection performance.

threshold of 0.05 achieves the best accuracy under both topologies, with both overly strict and overly lenient thresholds degrading performance. Three GMM components yield the highest graph classification accuracy, as fewer components fail to capture causal pattern diversity while more introduce redundancy. The optimal performance under both topologies is achieved at $K = 3$, where a smaller K fails to model long-range temporal causal dependencies of leak waves and a larger K introduces noise, making $K = 3$ the optimal trade-off between modeling depth and estimation stability. In summary, DCGN demonstrates notable robustness to optimizer selection and signal length, while appropriate configuration of causal prior parameters is critical to ensuring graph classification performance.

3.3. Comparative analysis with baseline methods

This section presents a systematic comparative evaluation of the proposed DCGN against six baseline methods on the leak detection task under a water supply flow rate of 0.18 L/s in Branched and Looped pipeline networks. The baseline methods span three categories. The first category comprises basic graph neural networks, including GCN with spectral-domain graph convolution and GAT with attention mechanisms. The second category encompasses advanced non-causal graph neural networks, including ATSGNN integrating spatiotemporal graph convolution with attention modules and MAGNet designed for acoustic emission signals in marine pipelines. The third category

Table 5

Accuracy statistics and post-hoc pairwise comparisons of all methods on leak detection under Branched and Looped pipeline topologies.

Topology	Baseline	Mean	Std	95% CI lower	95% CI upper	Mean Diff.	Adj. p	Cohen's d	Hedges' g
Branched	GCN	0.659	0.029	0.623	0.695	0.304	<0.001	13.637	12.317
Branched	GAT	0.815	0.007	0.806	0.823	0.149	<0.001	15.325	13.842
Branched	ATSGNN	0.860	0.006	0.853	0.867	0.103	<0.001	11.070	9.998
Branched	DCGN	0.963	0.012	0.948	0.978	–	–	–	–
Branched	MAGNet	0.927	0.020	0.902	0.951	0.037	0.011	2.228	2.013
Branched	CIGCN	0.924	0.022	0.897	0.951	0.039	0.011	2.219	2.004
Branched	GCT-GNN	0.923	0.016	0.903	0.944	0.040	0.004	2.785	2.516
Looped	GCN	0.726	0.015	0.707	0.745	0.240	<0.001	14.753	13.325
Looped	GAT	0.815	0.007	0.806	0.823	0.151	<0.001	11.559	10.441
Looped	ATSGNN	0.824	0.011	0.811	0.837	0.142	<0.001	9.886	8.929
Looped	DCGN	0.966	0.017	0.945	0.987	–	–	–	–
Looped	MAGNet	0.935	0.007	0.927	0.944	0.030	0.007	2.338	2.111
Looped	CIGCN	0.918	0.019	0.894	0.941	0.048	0.003	2.680	2.421
Looped	GCT-GNN	0.920	0.015	0.902	0.938	0.046	0.003	2.871	2.593

consists of causal inference-based graph neural networks, including GCT-GNN based on Granger causality tests and CIGCN achieving causal decoupling through structural causal models. All baseline methods are reproduced based on their original publications and implemented in PyTorch to ensure a consistent experimental environment. To ensure fair comparison, the hyperparameters of each baseline are tuned independently via grid search on the validation folds of the same five-fold cross-validation splits used for DCGN, and the configuration yielding the highest mean validation accuracy is selected as the final setting. All methods are evaluated with five-fold cross-validation using independent random seeds per fold to ensure statistical reliability.

As shown in Fig. 3, DCGN achieves the best performance under both pipeline topologies. Subfigures (a) and (b) present the accuracy and 95% confidence intervals of each method under Branched and Looped topologies, respectively, where DCGN reaches 96.32% and 96.60% and exhibits the narrowest error bars, indicating superior cross-fold stability. The one-way ANOVA results in subfigure (c) yield an F-value of 171.709 with degrees of freedom (6, 28), $p < 0.001$, and partial η^2 of 0.974, confirming highly significant inter-group differences. Furthermore, Table 5 presents post-hoc pairwise comparisons under Bonferroni correction, where all mean differences between DCGN and baseline methods reach statistical significance and Cohen's d effect sizes are consistently greater than 2, further corroborating the robust superiority of DCGN.

Regarding class-level detection reliability, subfigures (d) and (e) present the confusion matrices along with false alarm rates and missed detection rates of DCGN under Branched and Looped topologies, respectively. DCGN maintains per-class recall rates above 95% across all leak categories, with false alarm rates of 0.92% and 0.85% and missed detection rates of 3.68% and 3.40% under the two topologies, fully validating its balanced and reliable detection capability in engineering safety scenarios.

3.4. Robustness analysis

In practical pipeline monitoring system deployments, two major challenges arise, namely newly commissioned systems lack historical fault data leading to sample scarcity, and field sensor signals are contaminated by environmental noise and electromagnetic interference. To validate DCGN's engineering applicability under these real-world constraints, this section evaluates its robustness from two dimensions, including limited samples and multi-source noise interference. For data-scarce scenarios, training sets comprising 10% to 40% of samples were randomly selected from the complete dataset, with the remaining data serving as the test set. For noise interference scenarios, three typical disturbances were designed, including Gaussian white noise at SNR 5–30 dB superimposed on normalized signals to simulate environmental random disturbances, pink noise at the same SNR range injected to assess colored noise robustness, and salt-and-pepper noise at probabilities 0.01–0.05 applied to simulate abrupt sensor anomalies.

As illustrated in Fig. 4, DCGN exhibits exceptional robustness and reliability under conditions of limited training data and diverse noise interference. Fig. 4(a) reveals that DCGN achieves accuracy of 81.75% and 80.37% under Branched and Looped topologies, respectively, using merely 10% of training samples. As the training ratio increases to 40%, performance steadily improves to 98.37% and 98.24%, indicating that DCGN can train highly effective leak detection models even in small-sample scenarios. Furthermore, Fig. 4(b) and (c) demonstrate that DCGN possesses favorable noise resistance capability. Under extremely low SNR conditions of 5 dB, accuracy reaches 86.53% and 86.12% in Gaussian noise scenarios, and 88.98% and 87.76% in colored noise scenarios. As SNR increases to 30 dB, accuracy exceeds 97% under both topologies, significantly enhancing the reliability of pipeline health monitoring. Concurrently, Fig. 4(d) demonstrates that even under salt-and-pepper noise interference with a noise probability of 0.05, DCGN maintains accuracy of 93.88% and 93.74%, exhibiting robustness against abrupt anomalous points. In summary, through causal prior constraints and sparse skeleton learning, DCGN maintains robust and reliable pipeline leak detection performance under both sample-limited and strong noise interference scenarios.

3.5. Ablation study

To verify the effectiveness of each core component in DCGN, this section designs four groups of ablation experiments to respectively evaluate the contributions of the causal node encoder, causal skeleton learner, dynamic causal mechanism module, and causal inference module to pipeline leak detection performance. Specifically, EXP1 removes the causal node encoder and directly employs raw sensor data as graph node features; EXP2 removes the causal skeleton learner and replaces the learnable sparse causal skeleton with a fixed fully connected adjacency matrix; EXP3 removes the dynamic causal mechanism module and utilizes only the static adjacency matrix while neglecting the time-varying causal evolution process; EXP4 removes the causal inference module and employs GRU to replace the Transformer encoder for temporal reasoning. All ablation experiments were conducted with five-fold cross-validation under both Branched and Looped pipeline topology configurations, with evaluation metrics encompassing accuracy and F1-score.

As presented in Table 6, the complete DCGN model achieves accuracy of 96.32% and 96.60% under Branched and Looped topologies, respectively, significantly outperforming all ablation variants. EXP1 results reveal that removing the causal node encoder degrades performance to 82.86% and 80.41%, respectively, indicating that CNE effectively extracts robust pipeline state embeddings from raw waveforms through its parameter-shared one-dimensional convolutional encoder, thereby providing high-quality node representations for subsequent causal reasoning. Furthermore, EXP2 demonstrates the critical role of the causal skeleton learner, as the fixed fully connected adjacency

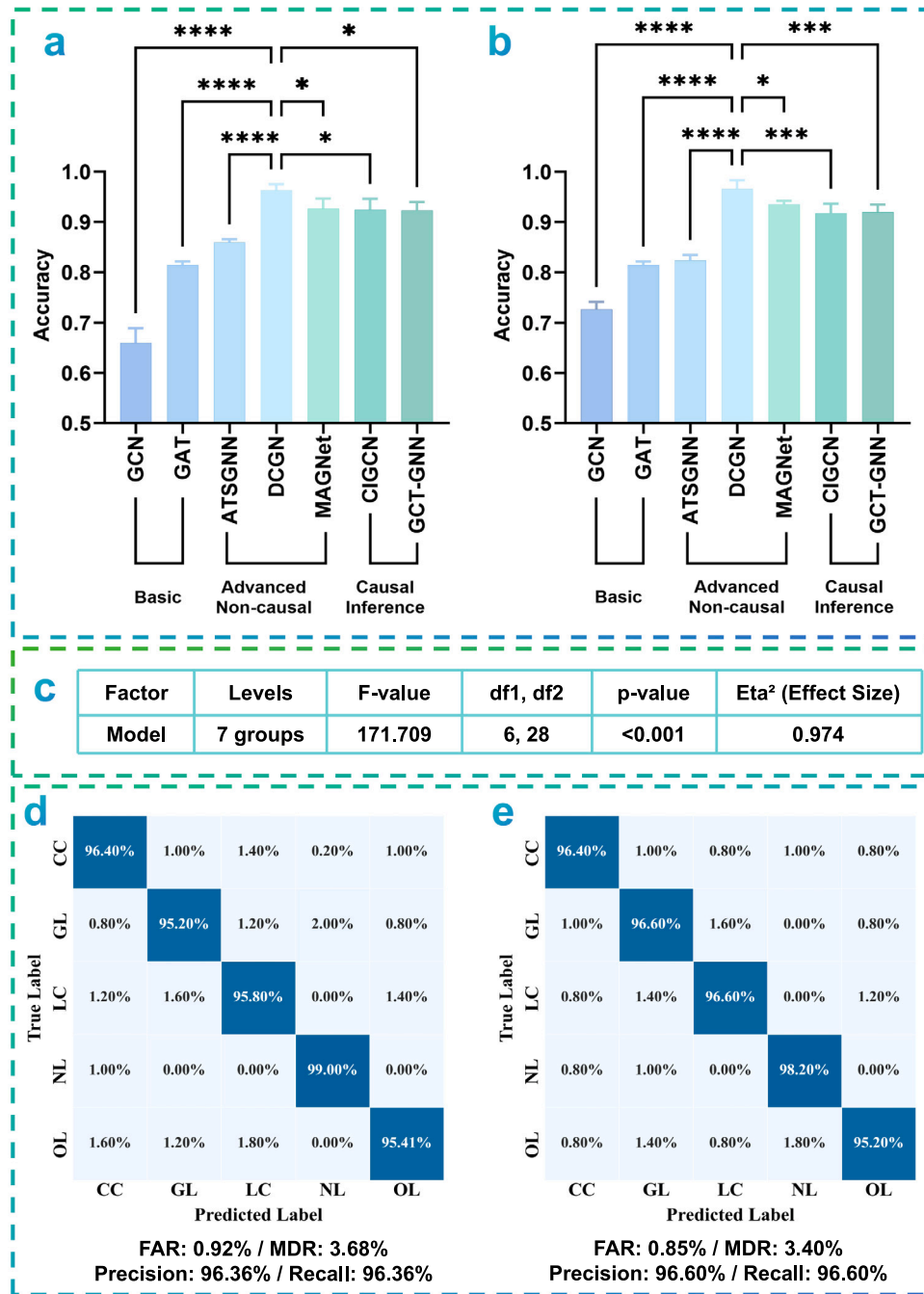


Fig. 3. Performance comparison of DCGN against baseline methods on pipeline leak detection. Subfigure (a) presents accuracy with 95% confidence intervals under the Branched topology across seven methods grouped into three categories. Subfigure (b) presents the corresponding results under the looped topology. Subfigure (c) summarizes the one-way ANOVA design and global test statistics across all seven model groups. Subfigure (d) shows the per-class confusion matrix, false alarm rate, and missed detection rate of DCGN under the Branched topology. Subfigure (e) shows the corresponding confusion matrix and reliability metrics of DCGN under the Looped topology.

matrix results in accuracy degradation to 83.27% and 84.08%, illustrating that the causal skeleton learned by CSL through sparsity and connectivity constraints can effectively eliminate spurious connections while preserving genuine physical causal pathways. Concurrently, EXP3 results indicate that the dynamic causal mechanism module is crucial, as removing DCM reduces accuracy to 89.39% and 88.98%, validating that the PCMI prior-based dynamic adjacency matrix generation mechanism precisely models the nonlinear propagation and attenuation processes of leak waves across the temporal dimension.

Finally, EXP4 demonstrates that the Transformer encoder in the causal inference module exhibits significant advantages over GRU, as removing CIR leads to accuracy degradation to 91.02% and 88.57%. This is attributable to the self-attention mechanism's capability to capture long-range causal dependencies across global temporal spans, avoiding the information attenuation issue inherent in sequential models. In summary, ablation experiments thoroughly substantiate that each component of DCGN makes an indispensable contribution to pipeline leak detection performance.

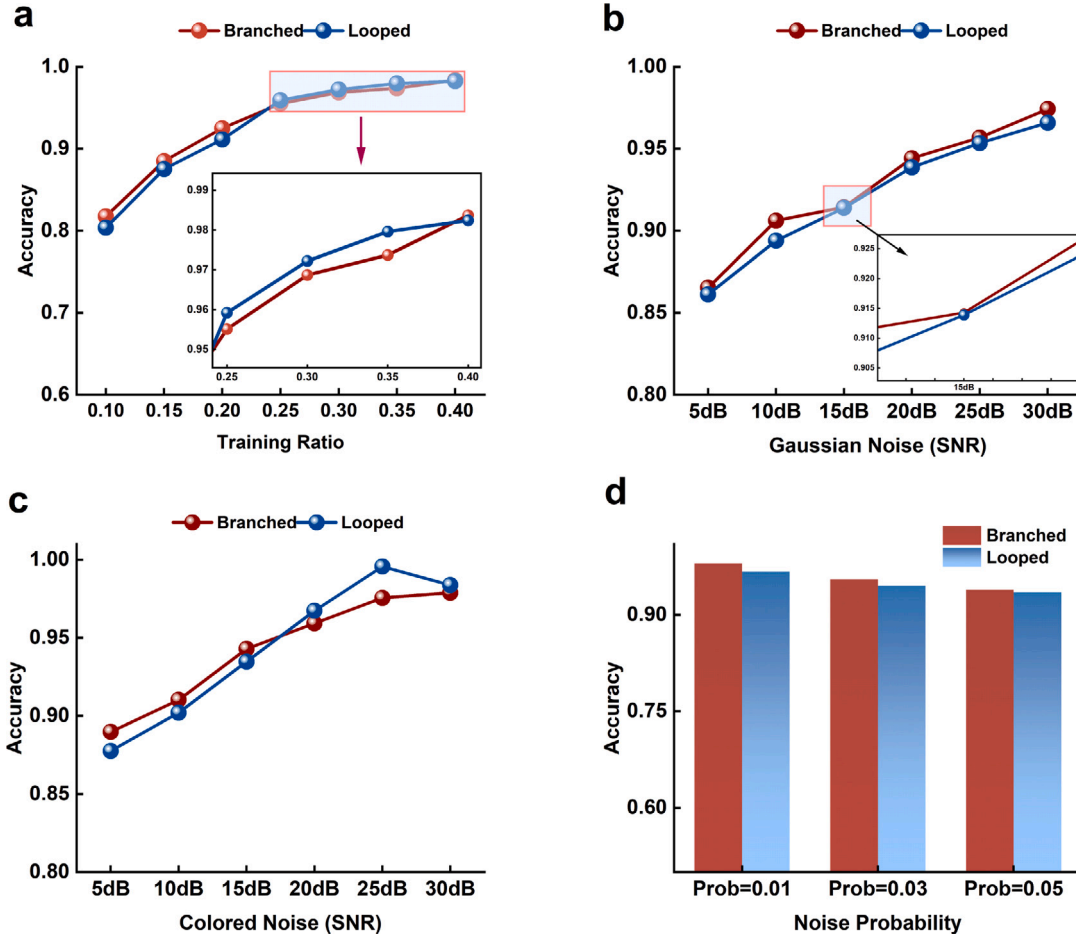


Fig. 4. Robustness analysis of DCGN under limited sample size and various noise conditions. Subfigure (a) shows performance with varying training ratios. Subfigure (b) shows robustness against Gaussian white noise at different SNR levels. Subfigure (c) shows robustness against colored noise at different SNR levels. Subfigure (d) shows robustness against salt-and-pepper noise at different noise probabilities.

Table 6

Ablation study results on Branched and Looped pipeline topologies.

Experiment	Branched accuracy	Branched F1-Score	Looped accuracy	Looped F1-Score
Full DCGN	96.32%	96.21%	96.60%	96.51%
w/o CNE	82.86%	82.86%	80.41%	80.86%
w/o CSL	83.27%	82.05%	84.08%	83.86%
w/o DCM	89.39%	89.17%	88.98%	88.82%
w/o CIR	91.02%	90.77%	88.57%	88.90%

Table 7

Computational overhead of full DCGN versus static graph variant.

Model	Parameters	Training time/ Epoch (s)	Inference latency/ Sample (ms)
w/o DCM	53,269	0.68 ± 0.07	4.47 ± 4.55
Full DCGN	327,723	0.92 ± 0.05	4.68 ± 1.95
Overhead	+274,094	+0.24	+0.21

To further evaluate the trade-off between the computational cost of dynamic causal graph generation and the associated performance gain, Table 7 presents the computational overhead of full DCGN against its static graph variant. The PCMI prior computation is performed offline over the training set only once prior to training, with complexity $O(N \cdot n^2 \cdot K)$ at $n = 4$ and $K = 3$, rendering this step negligible in overall cost. Compared with the static graph variant, full DCGN incurs

an additional 0.24 s per training epoch, 0.21 ms per inference sample, and 274,094 additional parameters. In summary, the marginal computational overhead introduced by dynamic causal graph generation is justified by the substantial performance gains reported in Table 6, demonstrating practical engineering feasibility.

3.6. Causal graph structure validation

To further assess the physical credibility of the graph structure learned by DCGN, this section examines three aspects including structural comparison, cross-fold stability, and sensor intervention, with results presented in Fig. 5. The physical ground truth is constructed from the actual pipe segment connections. As shown in subfigure (a), A1 serves as the hub node and is directly connected to P1, A2, and P2; A2 and P2 are directly connected through the bottom horizontal pipe; and neither P1 nor P2 has a direct connection to the other, with each relaying only through A1. For structural comparison, node-level saliency is derived through sensor ablation perturbation, and edge importance between any two nodes is quantified as the geometric mean of their saliency scores. Subfigure (b) shows that A1 obtains the highest saliency, reflecting its dominant role as the hub node of the network. Subfigure (c) shows that in the five-fold mean edge importance heatmap, the connections between A1 and P1 and between A1 and A2 carry the highest weights, which is consistent with the physical fact that these sensor pairs communicate through directly connected main-trunk segments. As shown in subfigure (d), binarising the learned

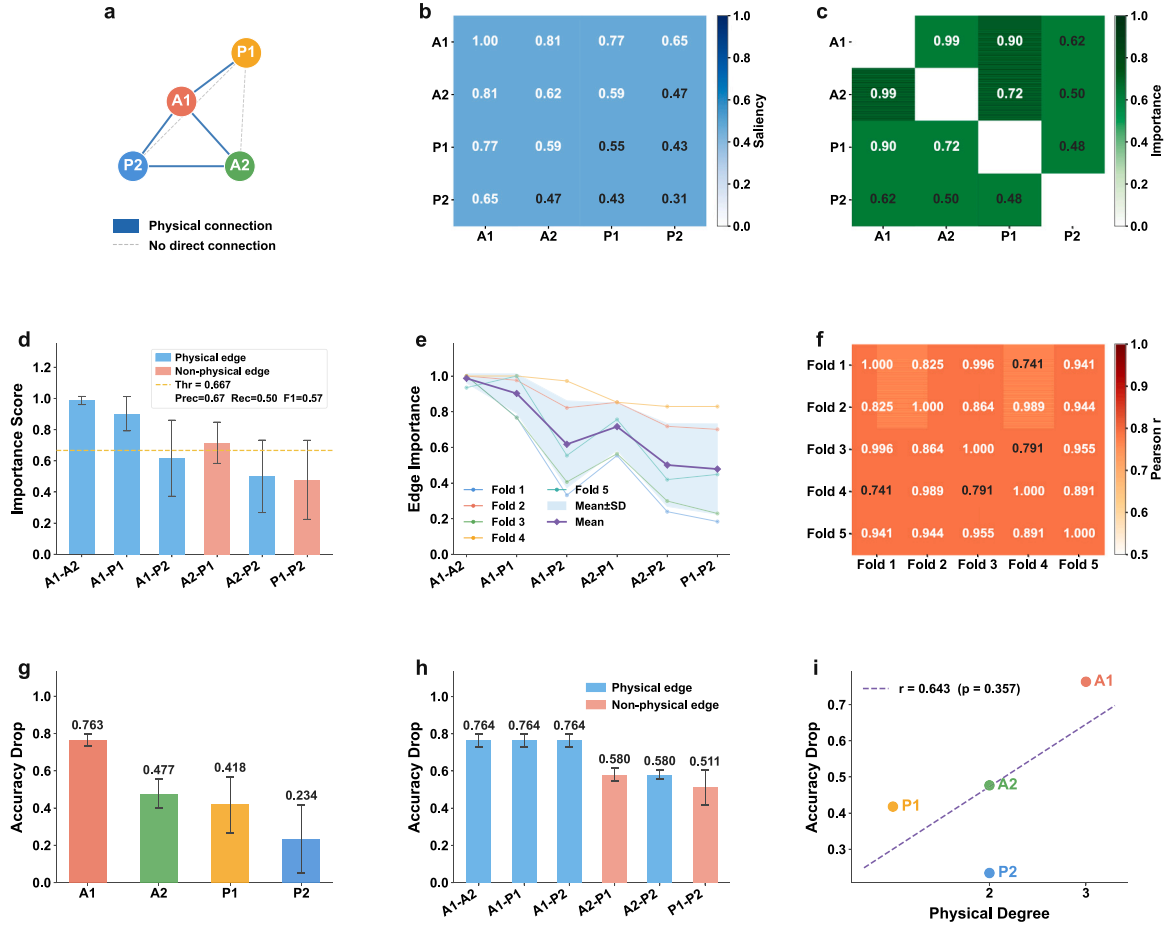


Fig. 5. Causal graph structure validation on the Branched topology at 0.18 L/s with sensors A1, A2, P1, and P2. Subfigure (a) shows the physical pipeline topology with known direct pipe connections as ground truth, where A1 serves as the hub node connected to P1, A2, and P2. Subfigure (b) shows the node-level saliency heatmap derived from sensor ablation. Subfigure (c) shows the five-fold mean learned edge importance heatmap. Subfigure (d) shows per-edge importance scores for physical and non-physical edges with the binarization threshold and structural alignment metrics. Subfigure (e) shows edge importance trajectories across all five folds with mean and standard deviation band. Subfigure (f) shows the cross-fold Pearson correlation matrix of edge-importance vectors. Subfigure (g) shows accuracy drop after sensor node masking for each node. Subfigure (h) shows accuracy drop after simultaneous masking of both endpoint node features for each edge. Subfigure (i) shows the scatter plot of physical node degree versus accuracy drop with a fitted regression line.

graph at the 50th-percentile threshold yields precision, recall, and F1 values of 0.667, 0.500, and 0.571 respectively. The mean importance of physical edges at 0.752 exceeds that of non-physical edges at 0.597, indicating that the overall distribution of edge importance aligns with the physical topology and that DCGN possesses a degree of physically aligned edge selection capability.

Regarding cross-fold stability, subfigures (e) and (f) show that the mean inter-fold Pearson correlation of edge-importance vectors reaches 0.8937, with a minimum of 0.7413, indicating that the learned graph structure remains highly consistent across different data splits. This consistency originates from the continuous guidance of PCMI prior constraints on causal skeleton learning, which effectively suppresses structural noise introduced by random initialization. For sensor intervention experiments, subfigure (g) shows that masking the features of each sensor node in turn causes the largest accuracy drop at A1 reaching 0.763, intermediate drops at A2 and P1 at 0.477 and 0.418 respectively, and the smallest at P2 at 0.234. This ordering closely matches the physical topological degree of each node, as nodes with higher degree carry more information propagation paths and their absence thus exerts a larger impact on model inference. Subfigure (i) further reveals a Pearson correlation of $r = 0.643$ between accuracy

drop and physical degree, corroborating that the internal information flow of DCGN has a meaningful correspondence with the node centrality of the true pipeline topology. Furthermore, subfigure (h) shows that simultaneously masking the features of both endpoint nodes for each edge yields a mean accuracy drop of 0.764 for physical edges, which consistently exceeds the values of 0.580 and 0.511 observed for non-physical edges, providing edge-level corroboration of the physical credibility of the causal paths in the learned graph.

3.7. Generalization across flow regimes and topologies

The core value of causal modeling lies in its generalization under distribution shift, where flow regime and pipeline topology represent the most prominent distributional variables in practical pipeline monitoring. To this end, this section evaluates the generalization capability of DCGN from two perspectives, namely within-regime adaptability and cross-regime transferability, with results presented in Fig. 6. As shown in subfigures (a) and (b), DCGN consistently achieves accuracy above 95% when trained independently under three flow conditions, 0 L/s, 0.18 L/s, and 0.47 L/s, across both Branched and Looped topologies, with comparable performance across regimes, indicating uniform

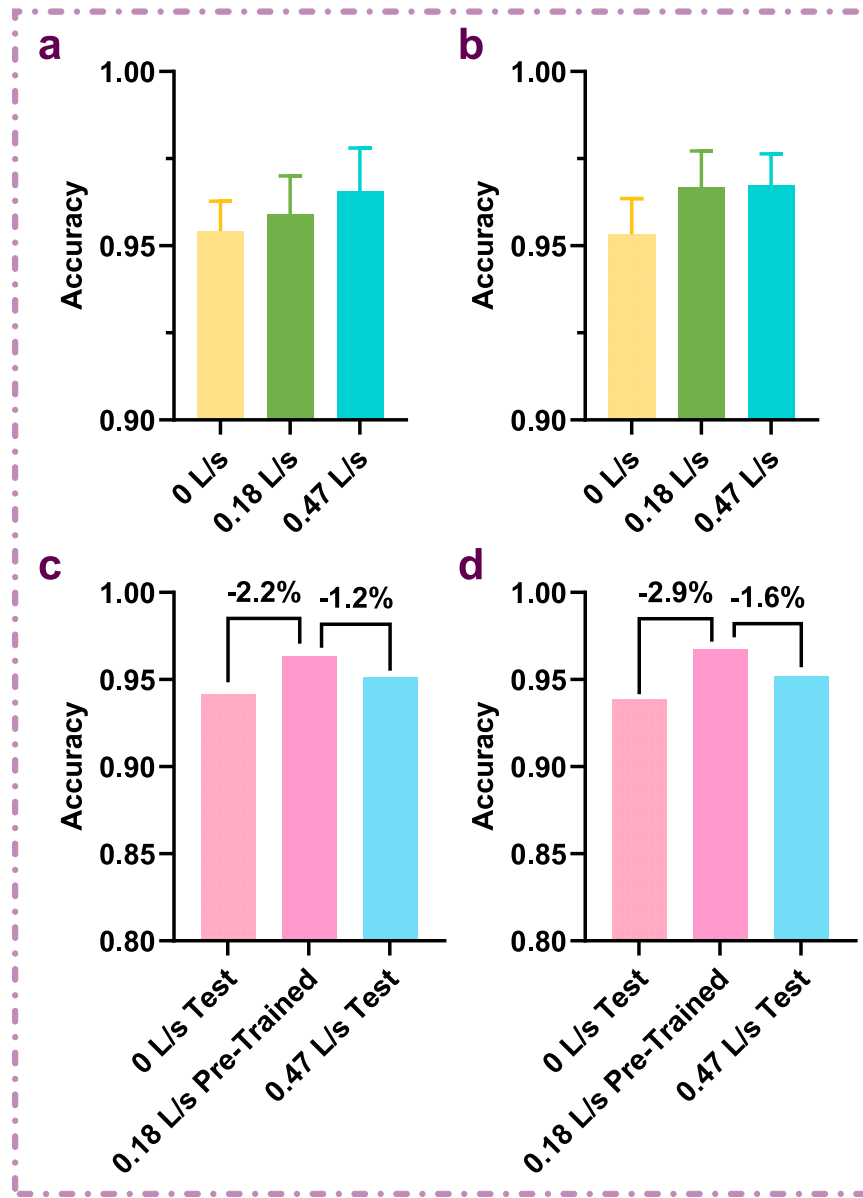


Fig. 6. DCGN accuracy under multiple flow regimes and cross-regime transfer. Subfigures (a) and (b) show within-regime performance under three flow conditions on Branched and looped topologies, respectively. Subfigures (c) and (d) show accuracy degradation under cross-topology transfer at 0.18 L/s, with subfigure (c) showing the Branched-trained model transferred to the Looped topology and subfigure (d) showing the looped-trained model transferred to the Branched topology.

adaptability to varying hydrodynamic conditions and confirming the cross-regime effectiveness of causal feature extraction. Furthermore, subfigures (c) and (d) examine the cross-topology transfer generalization capability of DCGN by applying models trained on one topology at 0.18 L/s to the other topology for inference, specifically transferring the model trained on the Branched topology to the Looped topology and transferring the model trained on the Looped topology to the Branched topology. The maximum performance degradation does not exceed 2.2% when tested on the Branched topology and 2.9% when tested on the Looped topology, demonstrating that the leak propagation pathways captured through causal skeleton learning exhibit structural invariance across pipeline topologies and thus enable the model to sustain high recognition accuracy under topological distribution shift, supporting flexible deployment across different pipeline configurations.

3.8. Discussion on practical reliability

The experimental results presented above validate the detection performance of DCGN under controlled benchmark conditions. However, advancing this framework toward practical industrial deployment requires addressing several reliability-level concerns.

First, regarding the reliability of sensor data sources and noise tolerance boundaries. In real-world deployments, sensors are inevitably subject to zero-point drift, calibration bias, and stochastic failures, all of which introduce systematic distortion at the signal level and may interfere with causal structure identification. DCGN incorporates inherent mechanisms to address this concern. CSL continuously eliminates weak spurious edges through sparsity and connectivity constraints during training, rendering the causal skeleton robust to mild signal distortion. The intervention training strategy in DCM randomly masks

a subset of causal pathways during each forward pass, compelling DCGN to learn redundant multi-path causal representations rather than relying on any single sensor channel, thereby sustaining diagnostic stability when sensor signal quality degrades. Regarding noise tolerance boundaries, the systematic robustness experiments presented earlier provide a quantitative assessment of performance degradation across varying noise intensities, demonstrating that DCGN maintains accuracy above 86% under extremely low SNR conditions of 5 dB and sustains accuracy above 93% under strong salt-and-pepper noise with a probability as high as 0.05, thereby revealing the upper noise bounds that the system can tolerate before diagnostic performance degrades significantly. These results indicate that sensor calibration errors or signal degradation must reach extreme levels before causing substantial performance decline, providing quantitative reference for sensor maintenance strategies in practical deployment.

Second, regarding the impact of sensor node failure on the overall system. Pipeline monitoring sensor networks exhibit series-system characteristics at the physical level, where the failure of a single node may disrupt the complete causal information chain. DCGN partially mitigates this concern through its sparse causal graph design, as the sparse skeleton ensures that each critical causal pathway is jointly supported by multiple sensor nodes, preventing a single-point failure from completely severing fault signal propagation routes. Nevertheless, the current framework remains limited in scenarios involving continuous sensor degradation or simultaneous multi-node failure, and future work will explore dynamic causal graph reconstruction mechanisms for node-failure scenarios to further enhance detection resilience under partial sensor absence.

4. Conclusion

This article proposes a Dynamic Causal Graph Network (DCGN) that integrates structural causal models with graph neural networks to disentangle genuine leak wave propagation paths from spurious correlations induced by confounding factors. DCGN employs PCMI-derived causal priors to constrain sparse skeleton learning and combines intervention-based dynamic causal mechanism generation to enhance pipeline monitoring reliability. Experimental results thoroughly demonstrate the superior performance of DCGN in pipeline leak detection tasks. Under Branched and Looped network topologies, DCGN achieves detection accuracy of 96.32% and 96.60%, respectively, significantly outperforming comparative methods such as GCN and MAG-Net, with ANOVA validating the statistical significance of performance differences. In robustness tests, DCGN exhibits exceptional noise resistance: achieving over 81% accuracy using merely 10% training samples, maintaining over 86% detection precision under extremely low SNR of 5 dB, and sustaining over 93% stable performance under strong salt-and-pepper noise interference. Ablation experiments demonstrate that removing either the causal skeleton learner or the dynamic causal mechanism module leads to substantial performance degradation, validating the effectiveness of causal inference mechanisms. Causal graph structure validation confirms that the learned skeleton aligns with the physical pipeline topology, and cross-topology transfer experiments demonstrate accuracy degradation within 2.9% when transferring between Branched and Looped topologies, further evidencing the distributional invariance of DCGN's causal representations. From a scientific perspective, this study advances the paradigm shift from correlation-based to causality-based modeling in graph neural networks by integrating structural causal models into pipeline structural health monitoring; from a societal perspective, the reliable low-false-alarm detection capability of DCGN facilitates the transition of critical infrastructure such as urban water supply and oil and gas transmission toward proactive preventive maintenance, holding practical significance for public safety and sustainable development. Nevertheless, the fixed-lag assumption of PCMI priors warrants further investigation under conditions of varying propagation speeds, while the latency and resource demands of real-time deployment remain to be assessed, and future work will address these directions.

CRedit authorship contribution statement

Yuxuan Zhang: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yuchen Lu:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dun Li:** Writing – review & editing, Supervision, Resources, Project administration, Methodology. **Sebastian Bader:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition. **Enrico Zio:** Writing – review & editing, Supervision, Resources, Project administration, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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